

Bharathraaj Nagarajan

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ML infrastructure engineer focused on distributed ML systems, scalable inference, and production AI infrastructure.

WORK EXPERIENCE

Media Sales Plus Inc., Buffalo, NY | Machine Learning Intern

Aug 2025 – Dec 2025

- Designed a **multi-tenant data ingestion and schema normalization pipeline** handling heterogeneous CSV/Excel formats across 10+ client datasets; automated schema detection, validation, and KPI computation to produce consistent structured outputs for downstream analytics.
- Implemented **time-series forecasting** (ARIMA, 12-month horizon) and **statistical anomaly detection** (IQR method) as independent, modular processing components; deployed via Streamlit on **Azure** for client-facing financial analytics.
- Built a **deterministic, rule-based analytics engine** for structured business queries (revenue trends, anomaly summaries, period comparisons); architected as a **modular, swappable component** designed to support future LLM integration without platform restructuring.
- Engineered **CI/CD automation** using GitHub Actions and Docker for automated, zero-downtime model deployment across cloud environments.

Tata Consultancy Services | ML Infrastructure Engineer – Model Serving & Distributed Pipelines

Oct 2021 – Apr 2024

- Engineered **real-time and batch model serving infrastructure** on Kubernetes for low-latency, high-throughput inference, achieving **99.9% production uptime** through containerized deployment, automated recovery, liveness probes, and observability instrumentation.
- Designed and optimized **feature engineering and ETL pipelines** using Spark (PySpark, Spark SQL), Kafka, HBase, and Kudu processing **100M+ records**, reducing ML training and inference latency by **~40%** through pipeline parallelization, partition strategy redesign, and Airflow-orchestrated scheduling.
- Owned **end-to-end ML model lifecycle** – data preparation pipelines, containerized deployment (Docker + Kubernetes), model versioning, A/B rollout, rollback, and real-time drift monitoring – for enterprise-scale financial ML workloads.
- Profiled and optimized **large-scale batch and streaming data pipelines** for ML training and inference, improving processing efficiency by **~47%** through distributed node parallelization, partition tuning, and systematic bottleneck identification.
- Collaborated with data science teams to translate model requirements into production-ready serving infrastructure, enabling **standardized self-service deployment workflows** for enterprise ML teams.

PROJECTS

CSE Pulse – Research Activity Intelligence Platform | Python, SQL, Groq/Llama 3.1, Docker, OpenShift, React | UB Research

- Building **production research data infrastructure** aggregating scholarly activity for **64 UB CSE faculty** across arXiv, DBLP, Semantic Scholar, IEEE, and RSS feeds; implemented entity resolution and deduplication to produce reliable structured datasets from noisy multi-source signals.
- Built **LLM inference pipeline** (Groq/Llama 3.1) classifying faculty research activity into grants, publications, collaborations, and research themes; automated departmental research digests and stakeholder analytics via **React dashboard**.
- Deploying on UB's **OpenShift Kubernetes cluster** under enterprise security constraints; containerizing services with Docker while satisfying non-root execution requirements.

Production ML Pipeline: End-to-End Training, Inference & Monitoring | Link

- Developed an **educational production-style ML pipeline** demonstrating reproducible workflows across data ingestion, feature engineering, model training (PyTorch), containerized serving (Docker + Kubernetes), experiment tracking and model registry (MLflow), CI/CD automation, and **drift monitoring**.

Distributed Edit Distance Benchmarking | Python, Apache Spark, Hadoop

- Built and benchmarked **distributed processing pipelines** on **100K+** datasets, achieving **20× speedup** over single-node baseline via Spark multiprocessing; analyzed throughput, latency, and compute efficiency tradeoffs across distributed node configurations.

AI-Powered Patient Scheduling Portal | Next.js, TypeScript, GPT-4o, AWS EC2, Twilio, Nginx | Link

- Built and deployed an **LLM-powered scheduling application**: GPT-4o handles conversation and intent, while all booking state is managed deterministically by application code – a clean boundary that prevents LLM hallucinations from corrupting booking data; deployed live on **AWS EC2**.

TECHNICAL SKILLS

Distributed Systems & Infrastructure: Kubernetes, Docker, Apache Spark (PySpark, Spark SQL), Kafka, Hadoop, HDFS, HBase, Kudu, Airflow, CI/CD (GitHub Actions), OpenShift

ML Infrastructure & MLOps: Model Serving, MLflow, Experiment Tracking, Model Registry, Drift Detection, Feature Engineering, Batch & Real-Time Inference, Model Versioning, Rollback

ML & Modeling: PyTorch, TensorFlow, Scikit-learn, Transformers, HuggingFace, Anomaly Detection, Time-Series Forecasting (ARIMA), NLP, Model Evaluation

Cloud Platforms: AWS (EC2, S3), Azure, Google Cloud Platform (BigQuery, Compute), Databricks, Snowflake

Programming: Python, SQL, System Design, Performance Debugging

EDUCATION

University at Buffalo, The State University of New York

Aug 2024 – Dec 2025

Masters in Data Sciences and Applications GPA: 3.875/4.0

Buffalo, NY

- Research Interests: ML Systems, Distributed Training Infrastructure, Inference Efficiency, and Empirical ML Research
- Coursework: Machine Learning, Data Intensive Computing, Distributed Systems, Database Management, Cybersecurity

Anna University – Sri Sivasubramaniya Nadar College of Engineering

Aug 2017 – May 2021

Bachelor of Engineering, Minor in Computer Science

Chennai, India